

Dalarna University

Beta black-box non-functional accessibility testing based on four personas

Version: Beta

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Project: Web Accessibility

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Introduction

This report presents the results of beta black-box non-functional accessibility testing performed on Dalarna University's website (www.du.se). The testing aimed to evaluate accessibility and usability in accordance with WCAG 2.1 AA standards using four representative personas: a user with visual difficulties, a user with motor difficulties, an inexperienced computer/phone user, and an active mobile user. The goal was to ensure that the website can be effectively used by individuals with diverse accessibility needs, especially regarding navigation, readability, and interaction.

1.1 Concepts and abbreviations

Test plan: Document outlining the scope, approach, environment, deliverables, and risk factors of testing activities (Hambling, 2010).

Test report: Summarized document of testing outcomes, results, and recommendations (Hambling, 2010).

Black-box testing: Testing approach where the internal workings of the application are not known to the tester.

Non-functional testing: Testing focused on quality attributes such as usability, accessibility, performance, and compatibility.

Accessibility testing: Verification that a product or service is usable by people with disabilities, ensuring compliance with WCAG 2.1 guidelines.

Error report: A structured record documenting deviations, issues, and actions taken.

Visual impairment: Reduced vision not corrected by standard glasses or contact lenses.

Motor difficulties: Conditions affecting fine motor control and interaction precision.

Inexperienced user: User with limited technical skills or understanding of navigation patterns.

Active mobile user: User primarily interacting via mobile devices using touch gestures.

Summary and Recommendations

The web accessibility evaluation demonstrated strong overall compliance with WCAG 2.1 AA standards. Out of sixteen executed test cases, fourteen passed successfully, while two revealed minor to moderate issues. These issues concern focus visibility during keyboard navigation and responsive rendering of PDF content on mobile devices.

Recommendations per persona:

- Visual difficulties (Vehid Sumic): Improve focus indicator visibility when tabbing through interactive elements. Ensure all menu icons and dropdowns provide clear focus highlights.
- Motor difficulties (Adrian Nasrat): Maintain consistent tab order and avoid focus traps.

Current implementation passed all tests.

- Inexperienced user (Anna Backolars): Search functionality and menu navigation are intuitive. Consider merging similar search results to reduce confusion.
- Active mobile user (Nengxi Peng): Replace or reformat static PDF documents into responsive HTML to ensure text readability after zooming or rotation.

Execution

Testing was conducted between October 6 and October 14, 2025, using multiple browsers and devices. All testers executed predefined test cases under realistic accessibility settings including screen readers, zoom levels up to 200%, and keyboard-only navigation. Minor technical challenges were encountered during mobile testing, primarily related to PDF responsiveness.

Result

Summary of test outcomes:

Total Test Cases	Executed	Passed	Failed
16	16	14	2

Errors Found

1. Missing focus highlight when navigating with keyboard (Vehid Sumic, Test Case 3) – Severity: S2 (Moderate)
2. PDF layout breaks on mobile rotation, text too small and distorted (Nengxi Peng, Test Case 4) – Severity: S3 (Minor)

Errors Remaining

Two issues remain unresolved. The focus visibility problem affects users relying on keyboard navigation and should be addressed as a priority. The mobile PDF readability issue can be resolved by converting static documents into HTML or responsive PDF format.

Summary of Test Activities

Testing activities were performed collaboratively by us representing four user personas. Each tester executed four test cases corresponding to their assigned accessibility scenario.

All test cases followed the same format including prerequisites, data, steps, expected and actual results, and pass/fail evaluation.

Experience

This beta accessibility testing process emphasized the importance of considering multiple user perspectives. It revealed how minor visibility or layout inconsistencies can impact inclusiveness. The collaboration between testers with different accessibility focus areas improved overall awareness of usability barriers.

References

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